

Artificial Potential Fields for Mobile Robot Navigation

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Abstract

This project implements artificial potential fields for two-dimensional mobile robot navigation. The target is modeled as an attractive potential minimum, while circular obstacles generate repulsive potentials within a finite influence distance. The robot evaluates the attractive and repulsive forces at its current position, sums them, normalizes the resulting vector, and advances in that direction using a discrete motion update. The main implementation uses a quadratic attractive potential, which produces an attractive force proportional to the distance from the goal. Conic and mixed attractive potentials are also analyzed as alternative formulations. The resulting scalar potential is represented as a three-dimensional surface and a contour map, while the negative potential gradient is visualized as a vector field. The experiment demonstrates obstacle avoidance without global path planning, but also reveals characteristic limitations of the method, including local equilibria and oscillation caused by normalized fixed-length motion steps near the goal.

1 Introduction

Artificial potential fields provide a compact method for reactive robot navigation. Instead of calculating a complete path in advance, the robot interprets its environment as an artificial energy landscape. The goal forms a low-potential region that attracts the robot, whereas obstacles form high-potential regions that repel it.

At every simulation step, the robot calculates the local force produced by this landscape. The attractive contribution points toward the goal and the repulsive contributions point away from nearby obstacles. The sum of these forces defines the instantaneous movement direction.

The method can be summarized as

$$\begin{aligned} &\text{goal attraction} + \text{obstacle repulsion} \\ &\quad \longrightarrow \text{local motion direction.} \end{aligned}$$

This project focuses on the quadratic attractive formulation. A second implementation demonstrates the conic,

or linear, attractive formulation, while a mixed formulation is discussed as a compromise between both models. The project also develops three complementary visualizations: the scalar potential surface, a two-dimensional contour map, and the vector field of normalized force directions.

2 Theoretical Background

2.1 Robot, Goal, and Circular Obstacles

The robot configuration is represented by the two-dimensional position

$$\mathbf{q} = \begin{bmatrix} x \\ y \end{bmatrix},$$

and the goal position by

$$\mathbf{q}_g = \begin{bmatrix} x_g \\ y_g \end{bmatrix}.$$

Each circular obstacle i is described by its center

$$\mathbf{c}_i = \begin{bmatrix} c_{x,i} \\ c_{y,i} \end{bmatrix}$$

and radius r_i . This representation allows the repulsive interaction to be based on the distance from the robot to the obstacle boundary rather than only to its center.

2.2 Potential and Force

Let $U(\mathbf{q})$ be a scalar potential defined over the robot workspace. The associated force is the negative gradient of that potential:

$$\mathbf{F}(\mathbf{q}) = -\nabla U(\mathbf{q}). \quad (1)$$

The gradient points in the direction of greatest potential increase. Therefore, its negative points in the direction of steepest decrease. The robot follows this descending direction locally at every iteration.

The total potential is decomposed into attractive and repulsive terms:

$$U_{\text{total}}(\mathbf{q}) = U_{\text{att}}(\mathbf{q}) + \sum_i U_{\text{rep},i}(\mathbf{q}). \quad (2)$$

Consequently,

$$\mathbf{F}_{\text{total}} = \mathbf{F}_{\text{att}} + \sum_i \mathbf{F}_{\text{rep},i}. \quad (3)$$

2.3 Quadratic Attractive Potential

The main implementation uses the quadratic attractive potential

$$U_{\text{att}}(\mathbf{q}) = \frac{1}{2} k_{\text{att}} \|\mathbf{q} - \mathbf{q}_g\|^2, \quad (4)$$

where $k_{\text{att}} > 0$ controls the strength of the attraction. The potential is always non-negative and reaches its unique minimum at the goal:

$$\mathbf{q} = \mathbf{q}_g \Rightarrow U_{\text{att}} = 0.$$

The expression is analogous to the potential energy of a linear spring. Its gradient is

$$\nabla U_{\text{att}} = k_{\text{att}}(\mathbf{q} - \mathbf{q}_g),$$

and therefore the attractive force is

$$\mathbf{F}_{\text{att}} = k_{\text{att}}(\mathbf{q}_g - \mathbf{q}). \quad (5)$$

The force points directly from the robot toward the goal. Its magnitude is

$$\|\mathbf{F}_{\text{att}}\| = k_{\text{att}} \|\mathbf{q}_g - \mathbf{q}\|,$$

so the unnormalized attractive force becomes stronger as the robot moves farther from the goal and weaker as it approaches the minimum.

2.4 Conic or Linear Attractive Potential

A conic attractive landscape can be defined as

$$U_{\text{att}}^{\text{lin}}(\mathbf{q}) = k_{\text{att}} d_g, \quad d_g = \|\mathbf{q} - \mathbf{q}_g\|. \quad (6)$$

For $d_g > 0$, the associated force is

$$\mathbf{F}_{\text{att}}^{\text{lin}} = k_{\text{att}} \frac{\mathbf{q}_g - \mathbf{q}}{\|\mathbf{q}_g - \mathbf{q}\|}. \quad (7)$$

Its magnitude is approximately constant and equal to k_{att} . Unlike the quadratic formulation, the attraction does not grow without bound as the distance increases. However, the conic potential is not smooth at the goal. The separate linear implementation therefore suppresses the attractive force inside a small neighborhood of the goal.

The linear demonstration also uses a different obstacle influence distance from the quadratic experiment. Its trajectory should consequently not be interpreted as a comparison caused exclusively by the attractive potential.

2.5 Mixed Attractive Potential

A mixed formulation combines the smooth behavior of the quadratic potential near the target with a bounded attractive force far from it. Let $d^* > 0$ be a switching distance. The potential can be defined as

$$U_{\text{att}}^{\text{mix}}(d_g) = \begin{cases} \frac{1}{2} k_{\text{att}} d_g^2, & d_g \leq d^*, \\ k_{\text{att}} d^* \left(d_g - \frac{1}{2} d^* \right), & d_g > d^*. \end{cases} \quad (8)$$

The corresponding attractive force is

$$\mathbf{F}_{\text{att}}^{\text{mix}} = \begin{cases} k_{\text{att}}(\mathbf{q}_g - \mathbf{q}), & d_g \leq d^*, \\ k_{\text{att}} d^* \frac{\mathbf{q}_g - \mathbf{q}}{d_g}, & d_g > d^*. \end{cases} \quad (9)$$

Thus, the force is proportional to distance near the goal and has bounded magnitude $k_{\text{att}} d^*$ farther away. This formulation is discussed theoretically but is not implemented in the supplied experiment.

Table 1: Comparison of attractive potential models.

Model	Potential	Force
Quadratic	d_g^2	$\propto d_g$
Conic	d_g	constant
Mixed	quadratic/linear	proportional/bounded

2.6 Distance to an Obstacle Boundary

For obstacle i , the distance from the robot to its center is

$$d_{\text{center},i} = \|\mathbf{q} - \mathbf{c}_i\|.$$

Because the obstacle has radius r_i , the relevant clearance is

$$d_i = \|\mathbf{q} - \mathbf{c}_i\| - r_i. \quad (10)$$

The sign of this quantity has a direct geometric meaning:

$$\begin{aligned} d_i > 0 &\Rightarrow \text{robot outside the obstacle,} \\ d_i = 0 &\Rightarrow \text{robot on the boundary,} \\ d_i < 0 &\Rightarrow \text{robot inside the obstacle.} \end{aligned}$$

2.7 Repulsive Potential

Repulsion is active only inside a finite influence distance ρ_0 . To avoid division by zero or excessively large numerical values, the clearance used in reciprocal terms is bounded from below:

$$\bar{d}_i = \max(d_i, \varepsilon). \quad (11)$$

The repulsive potential implemented for obstacle i is

$$U_{\text{rep},i}(\mathbf{q}) = \begin{cases} \frac{1}{2} k_{\text{rep}} \left(\frac{1}{\bar{d}_i} - \frac{1}{\rho_0} \right)^2, & d_i \leq \rho_0, \\ 0, & d_i > \rho_0. \end{cases} \quad (12)$$

At $d_i = \rho_0$, the active expression becomes zero. As the robot approaches the obstacle boundary, $1/\bar{d}_i$ grows rapidly and the potential forms a steep barrier.

2.8 Repulsive Force Derivation

Inside the influence region, define

$$s(d_i) = \frac{1}{d_i} - \frac{1}{\rho_0}.$$

Ignoring the numerical lower bound temporarily, differentiation gives

$$\frac{d}{dd_i} \left(\frac{1}{d_i} \right) = -\frac{1}{d_i^2}.$$

Therefore,

$$\frac{dU_{\text{rep},i}}{dd_i} = -k_{\text{rep}} \left(\frac{1}{d_i} - \frac{1}{\rho_0} \right) \frac{1}{d_i^2}.$$

The unit vector pointing from the obstacle center toward the robot is

$$\hat{\mathbf{u}}_i = \frac{\mathbf{q} - \mathbf{c}_i}{\|\mathbf{q} - \mathbf{c}_i\|}. \quad (13)$$

Using the safeguarded clearance from Equation (11), the implemented repulsive force is

$$\mathbf{F}_{\text{rep},i} = \begin{cases} k_{\text{rep}} \left(\frac{1}{\bar{d}_i} - \frac{1}{\rho_0} \right) \frac{1}{\bar{d}_i^2} \hat{\mathbf{u}}_i, & d_i \leq \rho_0, \\ \mathbf{0}, & d_i > \rho_0. \end{cases} \quad (14)$$

The factor $1/\bar{d}_i^2$ appears from differentiating the reciprocal distance. It makes the repulsion increase sharply near the obstacle. Contributions from all obstacles are added to form the total repulsive force.

2.9 Normalized Total Force and Robot Motion

After adding attractive and repulsive contributions, the experiment normalizes the total force:

$$\hat{\mathbf{F}}_{\text{total}} = \frac{\mathbf{F}_{\text{total}}}{\|\mathbf{F}_{\text{total}}\|}. \quad (15)$$

If the total force is the zero vector, the normalization routine returns the zero vector. The robot position is then updated using

$$\mathbf{q}_{t+1} = \mathbf{q}_t + v \hat{\mathbf{F}}_{\text{total}} \Delta t. \quad (16)$$

Normalizing the force means that only its direction is used for motion. With $v = 0.2$ and $\Delta t = 1$, every nonzero update has length

$$\|\mathbf{q}_{t+1} - \mathbf{q}_t\| = 0.2.$$

This produces a constant translational step even when the unnormalized force becomes small near the goal.

2.10 Potential Landscape and Force Field

The scalar potential and the vector force field describe different aspects of the same navigation method:

- the three-dimensional surface shows the complete energy landscape;
- the contour map shows equal-potential curves viewed from above;
- the vector field shows the local direction of $-\nabla U_{\text{total}}$;
- the trajectory shows the positions actually visited by the robot.

In the vector-field visualization, every force vector is normalized before being plotted. The arrows therefore show direction rather than relative force magnitude. In the contour visualization, high potential values are clipped for display so that the obstacle peaks do not dominate the complete color scale.

2.11 Local Minima and Other Limitations

A local equilibrium occurs when

$$\mathbf{F}_{\text{total}}(\mathbf{q}) = \mathbf{0}, \quad \mathbf{q} \neq \mathbf{q}_g. \quad (17)$$

At such a point, attractive and repulsive forces cancel. Because the method uses only local information, it cannot determine that another route around the obstacles would eventually reach the goal. Narrow passages, symmetric obstacle arrangements, and strong repulsive gains can also produce oscillations or excessive detours.

The normalized motion update introduces an additional numerical limitation. The quadratic attractive force becomes small near the goal, but normalization removes this decrease in magnitude. Unless the fixed step happens to enter the tolerance region, the robot may pass across the goal and oscillate around it. A distance-dependent speed or an unnormalized force update would provide smoother convergence.

3 Experimental Configuration

The quadratic potential-field experiment begins at

$$\mathbf{q}_0 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \quad \mathbf{q}_g = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

Four circular obstacles are included in the environment. Each obstacle is defined by its center $\mathbf{c}_i$ and radius r_i , as shown in Table 2.

Table 2: Circular obstacle configuration.

Obstacle	Center $\mathbf{c}_i$	Radius r_i
O_1	$[3, 3]^T$	0.3
O_2	$[6, 1]^T$	0.5
O_3	$[3, 5]^T$	0.5
O_4	$[4, 4]^T$	0.1

The parameters used for the attractive and repulsive potential fields and for the discrete robot motion are listed in Table 3.

Table 3: Quadratic potential-field parameters.

Parameter	Symbol	Value
Attractive gain	k_{att}	2.5
Repulsive gain	k_{rep}	1.0
Influence distance	ρ_0	0.5
Minimum clearance	ε	0.08
Motion speed	v	0.2
Time step	Δt	1.0
Goal tolerance	d_{tol}	0.01
Maximum steps	N_{max}	10000

At each iteration, the attractive force and the repulsive contributions of all obstacles are calculated at the current robot position. The resulting total force is normalized and used in the discrete motion update given by Equation (16).

The simulation terminates when the distance between the robot and the goal satisfies

$$\|\mathbf{q}_t - \mathbf{q}_g\| \leq d_{\text{tol}},$$

or when the maximum number of iterations N_{max} is reached.

For the potential-field visualizations, the scalar potential is evaluated over a two-dimensional rectangular grid defined by

$$-1 \leq x \leq 8, \quad -1 \leq y \leq 7.$$

The grid contains 150 samples along each coordinate axis. A separate 25×25 grid is used for the vector-field representation, where the total force is evaluated and normalized at every grid point.

4 Results and Discussion

4.1 Three-Dimensional Potential Landscape

Figure 1 represents the total artificial potential as a three-dimensional surface. The horizontal axes correspond to the position coordinates x and y , while the vertical axis represents the total potential

$$U_{\text{total}}(\mathbf{q}) = U_{\text{att}}(\mathbf{q}) + \sum_i U_{\text{rep},i}(\mathbf{q}).$$

The quadratic attractive potential produces a broad bowl-shaped surface centered at the goal. Because

$$U_{\text{att}}(\mathbf{q}) = \frac{1}{2} k_{\text{att}} \|\mathbf{q} - \mathbf{q}_g\|^2,$$

the potential increases quadratically with the distance from the target. The goal therefore appears at the bottom of the attractive basin, where the attractive potential reaches its minimum value.

The repulsive potentials generate steep peaks around the circular obstacles. Their rapid growth is caused by the reciprocal-clearance term

$$\left(\frac{1}{\bar{d}_i} - \frac{1}{\rho_0} \right)^2,$$

which becomes large as the robot approaches an obstacle boundary. The protected clearance

$$\bar{d}_i = \max(d_i, \varepsilon)$$

prevents division by zero and limits the numerical value of the peaks, although they remain considerably steeper than the surrounding attractive surface.

The red curve represents the simulated trajectory evaluated on the potential surface. It begins at

$$\mathbf{q}_0 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

and descends toward the goal at

$$\mathbf{q}_g = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

For visual clarity, the trajectory is drawn with a small vertical offset above the surface. This offset changes

only the graphical representation and does not modify the calculated potential or the robot motion.

The path does not follow a perfectly straight descent. When the robot enters the repulsive influence region of an obstacle, the local slope of the total potential changes. The repulsive gradient modifies the direction of descent, causing the trajectory to move around the corresponding potential peak before continuing toward the goal.

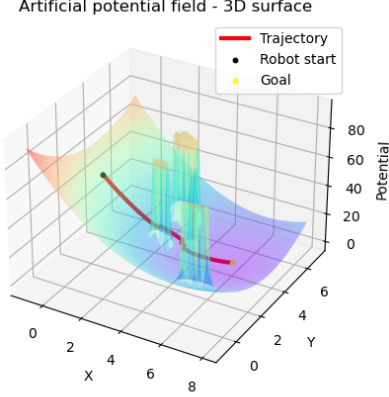


Figure 1: Three-dimensional total-potential surface. The quadratic attractive potential forms a broad basin around the goal, while the repulsive potentials generate steep peaks near the circular obstacles. The red curve shows the simulated robot trajectory.

4.2 Contour Map and Robot Trajectory

Figure 2 presents the same scalar potential from a top-down perspective. Every contour line connects points having the same total-potential value. The color distribution therefore provides a two-dimensional representation of the energy landscape shown in Figure 1.

Low-potential regions are concentrated around the goal, whereas higher potential values appear farther from the target and around the obstacles. The approximately circular contours surrounding the goal originate mainly from the quadratic attractive term. In the absence of obstacles, these contours would be concentric circles centered at $\mathbf{q}_g$.

Near an obstacle, the contour lines become strongly deformed and closely spaced. A high density of contour lines indicates a steep potential gradient. Consequently, these regions generate a strong change in the direction of the total force.

For visual readability, the displayed potential is limited to

$$0 \leq U_{\text{plot}} \leq 100.$$

This clipping affects only the color scale. The complete, unclipped potential is still used to calculate the forces and the robot trajectory. Without this graphical limit, the very large repulsive values close to the obstacle

boundaries would dominate the color map and make the attractive basin difficult to observe.

The circles drawn in the figure represent the physical obstacle boundaries. However, the repulsive force begins before the robot reaches these boundaries. Since the clearance is

$$d_i = \|\mathbf{q} - \mathbf{c}_i\| - r_i,$$

and repulsion is active when

$$d_i \leq \rho_0,$$

the effective distance of influence measured from the obstacle center is

$$R_{\text{influence},i} = r_i + \rho_0.$$

For the current experiment, the influence distances from the obstacle centers are therefore

$$\begin{aligned} R_{\text{influence},1} &= 0.3 + 0.5 = 0.8, \\ R_{\text{influence},2} &= 0.5 + 0.5 = 1.0, \\ R_{\text{influence},3} &= 0.5 + 0.5 = 1.0, \\ R_{\text{influence},4} &= 0.1 + 0.5 = 0.6. \end{aligned}$$

The robot initially moves approximately toward the goal under the dominant attractive force. Obstacle O_1 lies close to this direct route. As the robot enters its influence region, the repulsive contribution bends the trajectory upward and prevents collision with the obstacle.

After passing O_1 , the robot again turns toward the goal. A second local deviation appears near the smaller obstacle O_4 . Although its physical radius is only 0.1, it still generates repulsion within a clearance of $\rho_0 = 0.5$. The trajectory passes below this obstacle and subsequently returns toward the attractive minimum.

Obstacles O_2 and O_3 remain visible in the potential map, but they do not significantly modify the observed path because the trajectory does not enter their corresponding influence regions. The final robot marker lies within the goal tolerance and is therefore almost completely covered by the goal marker in the figure.

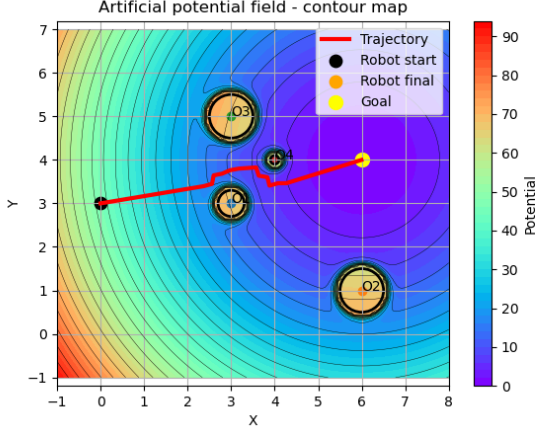


Figure 2: Contour map of the total potential and the resulting robot trajectory. Closely spaced contour lines indicate steep potential gradients near obstacle boundaries. Potential values are clipped only for graphical readability.

4.3 Normalized Force Vector Field

Figure 3 shows the direction of the total artificial force at different positions in the workspace. At every sampled point, the attractive and repulsive contributions are added:

$$\mathbf{F}_{\text{total}} = \mathbf{F}_{\text{att}} + \sum_i \mathbf{F}_{\text{rep},i}.$$

The plotted vector is then normalized:

$$\hat{\mathbf{F}}_{\text{total}} = \frac{\mathbf{F}_{\text{total}}}{\|\mathbf{F}_{\text{total}}\|}.$$

As a result, the arrows represent only the local motion direction. Their displayed lengths must not be interpreted as the magnitude of the actual force. A weak attractive force and a very strong repulsive force can appear with similar arrow lengths after normalization.

Far from all obstacles, the repulsive contribution is zero and the arrows point mainly toward the goal:

$$\mathbf{F}_{\text{total}} \approx \mathbf{F}_{\text{att}}.$$

The vector field therefore exhibits a general convergence pattern toward $\mathbf{q}_g$. This behavior is especially clear in the right-hand region of the map, where the arrows point toward the target from different directions.

Near obstacle boundaries, the repulsive force changes the orientation of the vectors. The arrows rotate away from the obstacle centers and create local flows around the circular regions. Around O_1 , the force field redirects the robot upward. Around O_4 , the field produces a second deflection that guides the robot below the obstacle.

The simulated trajectory follows these local force directions closely. Its shape can therefore be interpreted directly from the surrounding arrows:

local force direction $\longrightarrow$ next robot displacement.

Because the same normalized force is used in the discrete motion update,

$$\mathbf{q}_{t+1} = \mathbf{q}_t + v\hat{\mathbf{F}}_{\text{total}}\Delta t,$$

each nonzero displacement has the fixed length

$$v\Delta t = 0.2.$$

The small angular changes visible in the trajectory near the obstacles are therefore produced by the combination of a fixed motion step and rapidly changing repulsive directions.

Some arrows are also displayed inside the circular obstacles because the force field is evaluated over the complete rectangular grid. These vectors are a mathematical extension of the field used for visualization; positions inside the obstacles are not valid robot configurations and are not visited by the simulated trajectory.

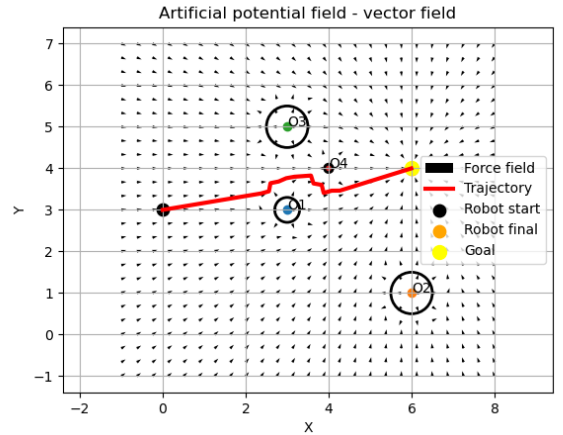


Figure 3: Normalized total-force vector field and simulated trajectory. Far from the obstacles, the arrows point mainly toward the goal. Near obstacle boundaries, the repulsive terms redirect the field away from the obstacle centers.

4.4 Combined Interpretation

The three representations describe different aspects of the same navigation process. The three-dimensional surface represents the artificial energy landscape, the contour map shows the spatial distribution of equal-potential regions, and the vector field shows the local direction of steepest potential decrease.

Their relationship can be summarized as

potential surface $\rightarrow$ energy landscape,
 contour map $\rightarrow$ top-down potential distribution,
 vector field $\rightarrow$ local movement direction,
 trajectory $\rightarrow$ positions actually visited.

The experiment demonstrates that the robot can reach the target while avoiding the obstacles without calculating a complete path in advance. At each iteration, it uses only the total force at its current position. Consequently, the final route emerges from repeated local decisions rather than from a global path-planning algorithm.

4.5 Influence of the Attractive Gain

The attractive gain determines the strength of the force directed toward the goal:

$$\mathbf{F}_{\text{att}} = k_{\text{att}} (\mathbf{q}_g - \mathbf{q}). \quad (18)$$

It also scales the quadratic attractive potential:

$$U_{\text{att}}(\mathbf{q}) = \frac{1}{2} k_{\text{att}} \|\mathbf{q} - \mathbf{q}_g\|^2. \quad (19)$$

Since the total force is normalized before updating the robot position, changing k_{att} does not modify the displacement length directly. Instead, it changes the balance between attraction and repulsion and therefore modifies the direction of motion.

4.5.1 Low Attractive Gain

For

$$k_{\text{att}} = 0.5,$$

the attractive contribution is relatively weak compared with the repulsive forces. The robot is strongly redirected near obstacles O_1 and O_4 , producing a larger deviation from the direct path and visible oscillations between both obstacles.

The three-dimensional surface also shows a shallower attractive basin. Therefore, the repulsive peaks have a stronger relative influence on the total potential.

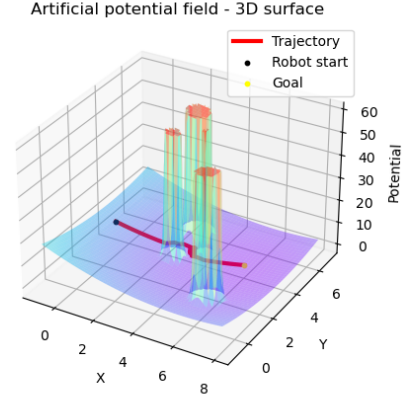


Figure 4: Potential surface for a low attractive gain, $k_{\text{att}} = 0.5$.

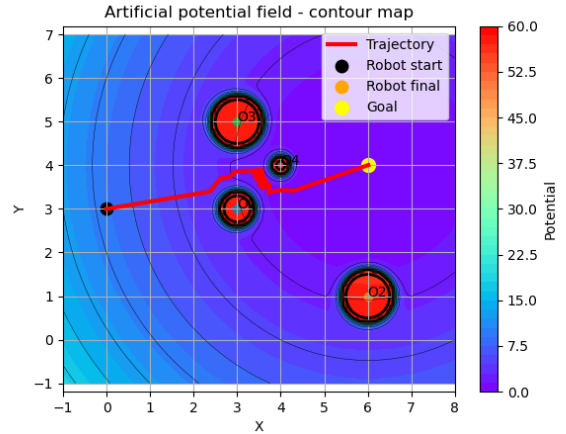


Figure 5: Contour map and robot trajectory for $k_{\text{att}} = 0.5$. The obstacle potentials have a strong influence relative to the shallow attractive basin.

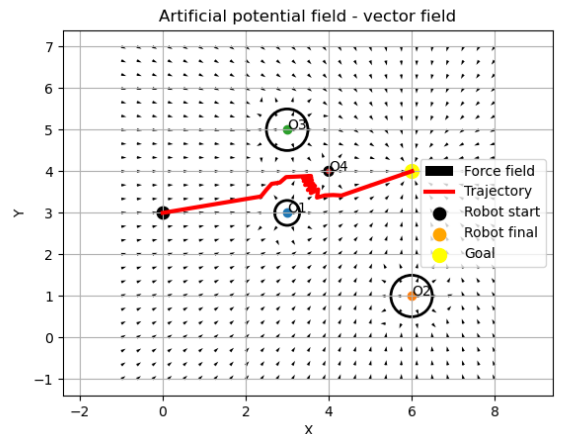


Figure 6: Normalized force field for $k_{\text{att}} = 0.5$. Strong changes in direction appear near O_1 and O_4 .

4.5.2 High Attractive Gain

For

$$k_{\text{att}} = 10,$$

the attractive force dominates a larger part of the workspace. The robot follows a smoother and more direct trajectory toward the goal, while the obstacles cause only local deviations.

The quadratic attractive potential also grows more rapidly with distance, producing a much steeper attractive basin. Nevertheless, an excessively large value of k_{att} reduces the relative influence of the repulsive forces and may cause the robot to pass too close to an obstacle.

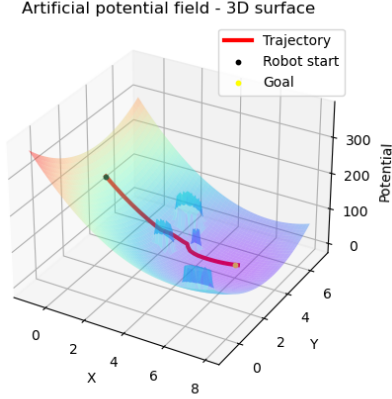


Figure 7: Potential surface for a high attractive gain, $k_{\text{att}} = 10$.

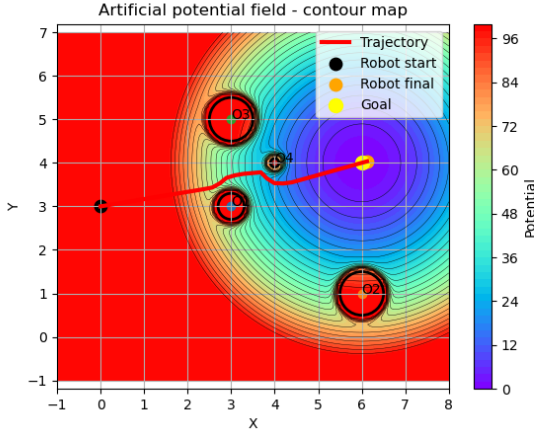


Figure 8: Contour map and robot trajectory for $k_{\text{att}} = 10$. The steep attractive basin dominates most of the workspace.

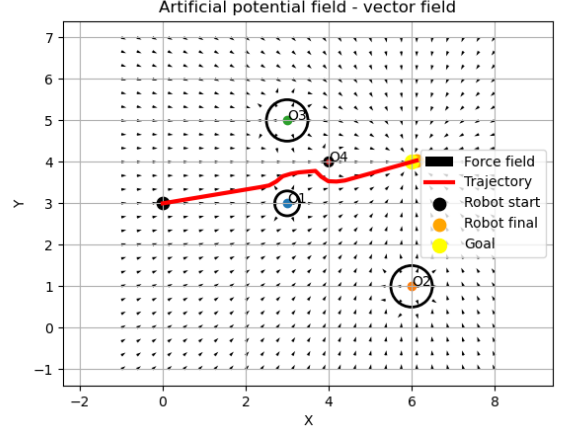


Figure 9: Normalized force field for $k_{\text{att}} = 10$. Most vectors point strongly toward the goal, except close to the obstacles.

The comparison can be summarized as

$$\begin{aligned} k_{\text{att}} \text{ small} &\Rightarrow \text{greater relative repulsive influence,} \\ k_{\text{att}} \text{ large} &\Rightarrow \text{greater attractive influence.} \end{aligned}$$

A suitable value must preserve sufficient attraction toward the goal without making the obstacle-avoidance contribution too weak.

4.6 Influence of the Repulsive Gain

The repulsive gain scales the obstacle potential and the corresponding repulsive force:

$$U_{\text{rep},i} = \frac{1}{2} k_{\text{rep}} \left(\frac{1}{d_i} - \frac{1}{\rho_0} \right)^2, \quad (20)$$

$$\mathbf{F}_{\text{rep},i} = k_{\text{rep}} \left(\frac{1}{d_i} - \frac{1}{\rho_0} \right) \frac{1}{d_i^2} \hat{\mathbf{u}}_i. \quad (21)$$

The repulsive gain was increased from its reference value $k_{\text{rep}} = 1$ to

$$k_{\text{rep}} = 5.$$

The influence distance remains unchanged at $\rho_0 = 0.5$. Therefore, increasing k_{rep} does not enlarge the region in which an obstacle is detected by the potential field. It only increases the strength of the repulsion inside that region.

The stronger repulsive contribution produces higher potential peaks around the obstacles. When the robot approaches O_1 and O_4 , the repulsive forces dominate the local direction of the total field and cause a stronger deviation from the direct path.

Because the total force is normalized before the position update, increasing k_{rep} does not increase the length of each motion step. Instead, it changes the direction of

motion by increasing the relative importance of obstacle avoidance.

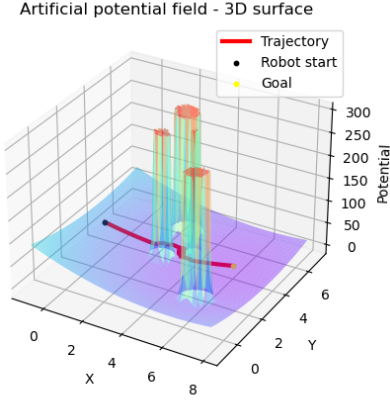


Figure 10: Potential surface for an increased repulsive gain, $k_{\text{rep}} = 5$. The obstacle peaks become more dominant relative to the attractive basin.

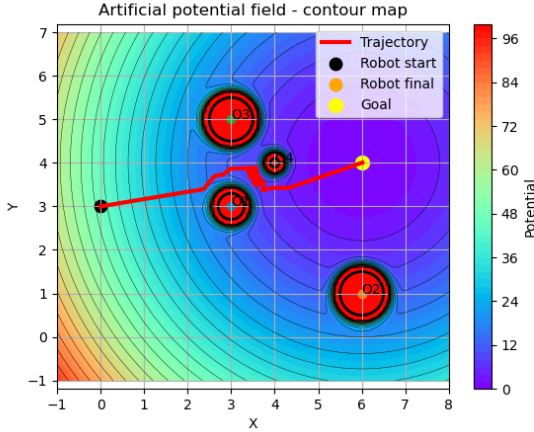


Figure 11: Contour map and robot trajectory for $k_{\text{rep}} = 5$. The stronger obstacle potentials deform the surrounding contours more noticeably.

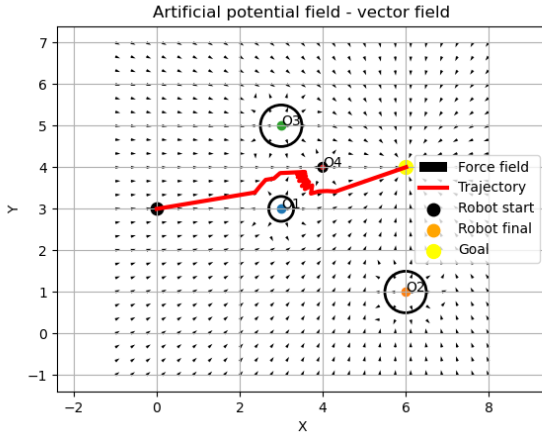


Figure 12: Normalized force field for $k_{\text{rep}} = 5$. Strong repulsive directions near O_1 and O_4 produce visible oscillations in the robot trajectory.

The experiment shows that a larger repulsive gain increases the safety response near obstacles, but an excessively large value can produce sharp deviations, oscillations, or unnecessarily long trajectories.

k_{rep} large $\Rightarrow$ stronger obstacle avoidance
and larger path deviations.

4.7 Influence of the Repulsive Influence Distance

The parameter ρ_0 defines the maximum clearance at which an obstacle generates a repulsive contribution:

$$\mathbf{F}_{\text{rep},i} = \mathbf{0}, \quad d_i > \rho_0. \quad (22)$$

Since d_i is measured from the obstacle boundary, the effective influence radius measured from its center is

$$R_{\text{influence},i} = r_i + \rho_0. \quad (23)$$

Therefore, ρ_0 mainly controls the spatial extent of obstacle avoidance. It also appears inside the repulsive-force expression and therefore modifies the force profile within the active region.

4.7.1 Small Influence Distance

For

$$\rho_0 = 0.1,$$

the repulsive force becomes active only when the robot is very close to an obstacle boundary. Most of the workspace is therefore dominated by the attractive force, and the robot follows an almost direct trajectory toward the goal.

This produces a short and smooth path, but the robot reacts late to obstacles and may pass dangerously close to their boundaries.

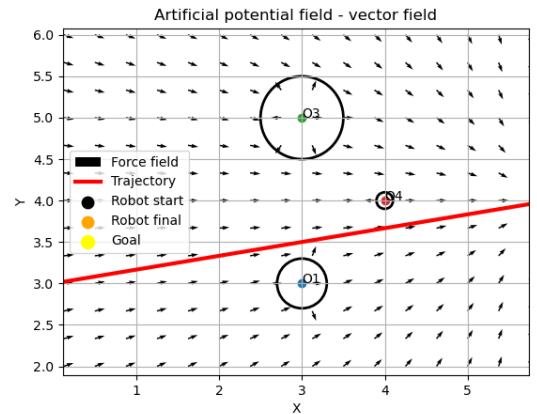


Figure 13: Normalized force field for a small influence distance, $\rho_0 = 0.1$. The obstacles affect only a narrow surrounding region, producing an almost direct trajectory toward the goal.

4.7.2 Large Influence Distance

For

$$\rho_0 = 10,$$

the obstacles influence almost the entire workspace. Repulsive contributions from several obstacles are therefore active simultaneously, even when the robot is far from their surfaces.

The resulting force field contains stronger directional distortions, and the trajectory shows visible oscillations and larger deviations. An excessively large value of ρ_0 can therefore make obstacle avoidance unnecessarily conservative and reduce progress toward the goal.

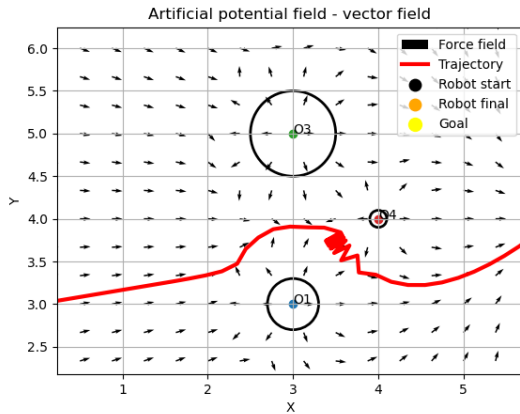


Figure 14: Normalized force field for a large influence distance, $\rho_0 = 10$. Multiple obstacles affect the robot simultaneously, producing stronger deviations and oscillations.

The influence of ρ_0 can be summarized as

- ρ_0 small $\Rightarrow$ late obstacle reaction,
- ρ_0 large $\Rightarrow$ wide and potentially excessive repulsion.

A suitable value should provide enough distance for safe obstacle avoidance without allowing distant obstacles to dominate the navigation field.

5 Conclusion

The experiment demonstrates how an attractive goal and circular obstacles can be combined within a single reactive navigation framework. The quadratic attractive potential creates a smooth goal-centered basin, while the finite-range repulsive potentials generate localized barriers around the obstacles. The negative gradient of the total potential provides the robot with a local motion direction without requiring the computation of a complete global path.

The three visualizations provide complementary interpretations of the same navigation process. The three-dimensional surface represents the artificial energy landscape, the contour map shows its spatial distribution,

and the vector field indicates the local direction of motion. The resulting trajectory emerges from repeatedly following the normalized total force.

The parameter study shows that the behavior of the method depends strongly on the balance between attraction and repulsion. A small k_{att} increases the relative influence of obstacles and may produce larger deviations or oscillations, whereas a large k_{att} produces a more direct trajectory but can weaken the obstacle-avoidance effect. Increasing k_{rep} strengthens the reaction near obstacles without changing the size of their influence regions. In contrast, ρ_0 directly determines how far from an obstacle the repulsive force becomes active.

The experiment also highlights an important difference between the potential-field equations and the numerical motion rule. Although the quadratic attractive force naturally decreases near the goal, normalizing the total force removes its magnitude information. With a fixed value of $v\Delta t$, the robot therefore moves using an approximately constant step length whenever the total force is nonzero. This can produce oscillations near obstacles or around the goal and may prevent a very small goal tolerance from being reached.

Artificial potential fields provide a simple and computationally efficient reactive navigation method. However, they rely only on local information and can suffer from local minima, oscillations, and excessive sensitivity to parameter selection. These limitations motivate later extensions such as adaptive step sizes, mixed attractive potentials, escape behaviors, or the combination of local potential fields with global path planning.

6 Project Task

Represent a mobile robot, one goal, and several circular obstacles in a two-dimensional workspace. Implement a quadratic attractive potential and a finite-range repulsive potential based on obstacle clearance. Derive the corresponding attractive and repulsive forces, sum all force contributions, normalize the total force, and update the robot position iteratively.

Stop the simulation when the robot enters a specified goal tolerance or when a maximum number of steps is reached. Visualize the total potential as a three-dimensional surface and contour map, draw the normalized force field, and overlay the resulting robot trajectory.

Finally, study the influence of the attractive gain k_{att} , the repulsive gain k_{rep} , and the repulsive influence distance ρ_0 . Compare the quadratic attractive formulation with the conic model and explain how a mixed potential can combine bounded far-field attraction with smooth near-goal convergence.